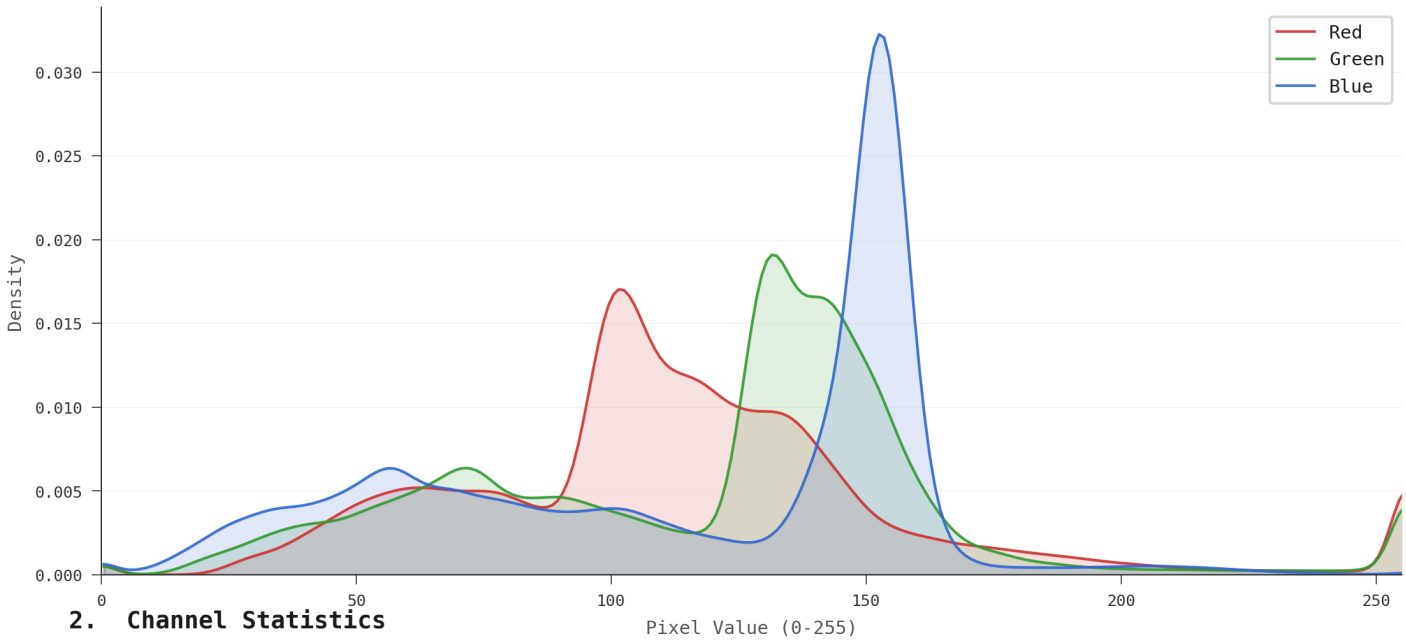
Reference Data Sheet: KODIM21

Subject: Portland Head Lighthouse

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	111.6135	41.6961	1738.5686	1.8853	0.8139	0	255
Green	118.0137	44.5940	1988.6265	0.5278	0.0245	0	255
Blue	114.5361	47.9700	2301.1177	-0.9321	-0.4663	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8957	0.6651
G	0.8957	1.0000	0.9155
B	0.6651	0.9155	1.0000

Pairwise Pearson Correlation Coefficients

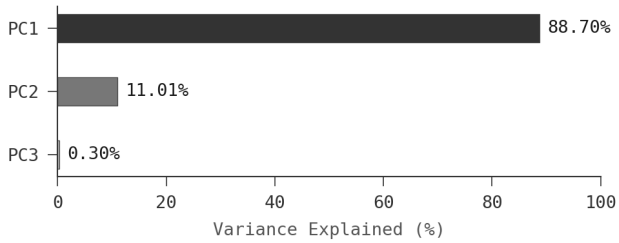
r(R-G)	=	**0.895717**
r(R-B)	=	**0.665054**
r(G-B)	=	**0.915487**
Avg |r|	=	**0.825419**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1738.5730} & 1665.4993 & 1330.2193 \\
1665.4993 & \mathbf{1988.6316} & 1958.3910 \\
1330.2193 & 1958.3910 & \mathbf{2301.1235}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM21

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5061	-0.6078	**-0.6119**	5346.8877
PC2	-0.7421	-0.0545	0.6680	663.5249
PC3	0.4394	-0.7922	0.4235	17.9155

6. Derived Metrics

Condition Number

298.45

(l1/l3) PC1/PC2: 8.06 | PC2/PC3: 37.04

Blue Channel Independence

13.0%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Blue

Loading: -0.6119 | Green-Blue coupled

Classification: Weakly One-Dimensional (PC1 = 88.70%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.945970	0.894270	0.920120
Green	0.951766	0.905705	0.928736
Blue	0.960706	0.924619	0.942662

8. Average Local Variance (3x3)

Channel	Value
Red	216.4076
Green	220.4911
Blue	197.4763

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 298.45 | PC1: 88.70%

Correlation: G–B pair leads at r = 0.915487 | R–G: 0.895717 | R–B: 0.665054

Blue Channel: 13.0% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.930506 | Avg local variance: 211.4583

Eigenvector: Green-Blue coupled | G (-0.6078), B (-0.6119) co-load | R offset (-0.5061)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."